

q_N reopened

the selected sequence has room; razor-overlap is vacuous-only; $|Z_{\text{loc}}|$ is the most provable coordinate

Stefan Hamann

August 29, 2026

Abstract

The edge-balance chain reduced the dual-resolvent edge half to $q_N < 1$. That terminal hole is reopened here on the *selected* sequence $a_k = 2^k$, with the arsenal accumulated since the lemma-hydra freeze: four candidate death coordinates, the COMPOSE⁻ implication, and the dyadic window $W_k = W^{\mathbb{R}}(2^k, \dots)$ as the remaining quantifier. On that sequence q_N stays far from 1 (minimum margin 0.617 versus 0.0195 on the old 42-rung ladder). Razor-overlap is *not* a global fourth equivalent coordinate: it separates living and dead only on the vacuous chart, and the C_{min} -mode is not the hole-Nyquist vector. The most provable remaining sublemma is a head-sum bound $|Z_{\text{loc}}(W_k)| \leq \frac{1}{2}$. No RH claim.

Status. Lemma 1 is **reduced** (a selected census with comfortable margin; cofinal $q_N < 1$ remains). Proposition 1 is a finite theorem (imported). Proposition 2 is a named refutation. Ausgang **SELECTED_MARGIN_GO** / **OVERLAP_VAC_ONLY** / **ZLOC_WINS** / **COFINAL_OPEN**. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The lemma

Lemma 1 (q_N on the selected sequence). $q_N < 1$ for all sufficiently large k of the selected sequence $W_k = W^{\mathbb{R}}(2^k, \dots)$.

Verdict: REDUZIERT as posted. Measured on $k = 3, 4, 5, 6, 7, 9$ (builder fallback at $k = 8$, $N_w = 5690$). The remaining analytic object is a bound on one of the four coordinates below, of which $|Z_{\text{loc}}|$ has the best proof profile on this sequence.

2 Dictionary and the four coordinates

Proposition 1 (terminal dictionary). On a canonical window, $q_N = \rho_{N-1}/(\frac{5}{7}) = \frac{7}{5}Z^2$. Hence $q_N < 1$ if and only if $|Z| < M$ with $M = \sqrt{5/7}$. Over $\mathbb{Q}$: $Z = \frac{1}{2}$ gives $q_N = \frac{7}{20}$. The death channel $|Z_{\text{loc}}| \geq M \iff |Z| \geq M \iff q_N > 1$ is a 181/181 census (r386). The chart identity $\text{sch} < 0$ on the living 42 and $\text{sch} > 0$ on the six terminal-dead χ -sprouts is a census (r417).

Four quantities were proposed as equivalent death/life coordinates:

$$q_N > 1 \iff |Z_{\text{loc}}| \geq M \iff \text{sch} > 0 \iff \text{razor-overlap } 10^{-4} \text{ vs } 10^{-8}.$$

The first three match on every world of this round (selected, core-42, dead χ). The fourth does not: see §4.

3 Selected census

The map $k \mapsto \text{kz}$ is the prime-power index of $a_k = 2^k$ in the von Mangoldt table: $k = 3, \dots, 9$ hit $\text{kz} = 5, 9, 17, 26, 43, 69, 116$. These are always genuine prime-power anchors, unlike the 42-rung ladder whose tightest rung ($\text{kz} = 16$, $q_N = 0.9805$) is *not* a high power of two.

k	kz	N_w	q_N	$ Z_{\text{loc}} $	R	L_1	sch	chart
3	5	72	0.383	0.487	0.60	0.31	−0.072	P1
4	9	184	0.214	0.157	1.40	0.40	−0.067	P1
5	17	96	0.023	0.227	1.17	0.39	−0.109	P1
6	26	364	0.00176	0.011	0.81	0.45	−0.069	VAC
7	43	839	0.00151	0.034	1.03	0.48	−0.055	VAC
8	69	5690	—	—	—	—	—	P1 [†]
9	116	1433	0.00070	0.076	1.28	0.63	−0.038	P1

[†] builder pin from r421 ($R_{\text{phibb}} = 0.0476$, $\text{den} = 1.569$); q_N not re-gated live.

q_N falls in k . It is *not* monotone in N_w ($k = 5$ has $N_w = 96 < 184 = N_w(k = 4)$). The selected abscissa is k , as in r421. Minimum margin to 1 on the live selected rungs is $1 - 0.383 = 0.617$, against 0.0195 on the 42-rung ladder (attained at the false-anchor $\text{kz} = 16$).

COMPOSE inputs on selected are comfortable: $S_F/D \leq 1.40 \ll R_0 = 4$, $L_1 \leq 0.63 \ll \Lambda = 3$, $|Z_{\text{loc}}| \leq 0.487 \ll Z'_0 = 21/25 = 0.84$, and M_3 stays under the r306 polylog floor on every live selected rung. The envelopes that were sharp on the living ladder (r386: 0.011 of air from Z_{loc} to M at $\chi_4 \text{kz} = 46$) are not sharp here.

4 Razor-overlap is vacuous-only

The r422 four-order split (living $\sim 10^{-8}$, dead $\sim 10^{-4}$) was measured on *vacuous* windows. It does not extend.

Proposition 2 (overlap is not global). *On the core 42, P1 living razor-share occupies $[8 \cdot 10^{-9}, 3.6 \cdot 10^{-2}]$ and overlaps the dead- χ range $[1.5 \cdot 10^{-4}, 2.4 \cdot 10^{-2}]$. Vacuous living core share is at most $2.9 \cdot 10^{-7}$; dead vacuous χ is at least $1.5 \cdot 10^{-4}$. The four-order separator is a vacuous-chart classifier, not a q_N -margin meter. Witness: $\text{kz} = 16$ (living, $q_N = 0.9805$, $|Z_{\text{loc}}| = 0.756$, $\text{sch} = -0.00151$) has razor-share $5.4 \cdot 10^{-9}$.*

The candidate mechanism was: when $q_N > 1$, border mass s walks into the bulk near-defect $\psi_{C_{\min}}$ (the $C_{\min}$ -mode, a near-Nyquist face of the hole Gram). On P1, $C_{\min} < 1$ and $\langle \psi_{C_{\min}}, v_{\text{Nyq}} \rangle \approx -0.386$ at $k_z = 9$ (the r410 number): s couples at share $\sim 10^{-2}$ *because the bulk already has a defect*, independently of $q_N < 1$. On the vacuous chart the candidate fails in the other direction:

- $\langle \psi_{C_{\min}}, v_{\text{Nyq}} \rangle$ is $O(10^{-2})$ ($k = 6$: -0.007 ; $k = 7$: $+0.006$). The $C_{\min}$ -mode is *not* the hole-Nyquist vector (r410: Nyquist is not a spectral atom).
- s still has Nyquist overlap $O(10^{-2})$, living or dead.
- Living vacuous orthogonality $|\langle s, \psi_{C_{\min}} \rangle|^2 / \|s\|^2 \sim 10^{-8}$ is orthogonality to an *implicit* near-threshold eigenmode of C , not an explicit oscillating source sum against $(-1)^j$.

Gate protocol. The 10^{-8} statement is not unconditional (it needs the C -spectrum), not PNT-elementary (no explicit phase), and would be RH-near if one tried to identify $\psi_{C_{\min}}$ with a prime-comb test function. The Nyquist rewrite is refuted. Overlap remains a sharp *vacuous* census, not the most provable coordinate.

5 Ranking

Coord.	Selected margin	Source	Proof profile
q_N	0.617 ($k=3$)	$\rho = F^2/h$	product recursion (the hydra); dictionary-equivalent to Z .
$ Z_{\text{loc}} $	0.358 to M ; 0.013 to $\frac{1}{2}$	explicit edge sum	COMPOSE ⁻ already a SATZ; remaining bound is a finite head-sum.
sch	floor ~ 0.038	Woodbury SATZ	vacuous reduction to $\varphi_{bb} < 0$ (r419); floor is a census.
overlap	4 orders, VAC only	$\langle s, \psi_{C_{\min}} \rangle$	implicit eigenmode; Nyquist rewrite dead; not global.

Winner: $|Z_{\text{loc}}|$. On selected it inherits the r378 implication and the envelopes are no longer sharp against M . The product q_N has more numerical room but the same recursive shape that froze the lane.

Lemma 2 (selected head bound, census). *On the MAIN selected windows W_k for $k \in \{3, 4, 5, 6, 7, 9\}$, $|Z_{\text{loc}}(W_k)| \leq \frac{1}{2} < M$. Via COMPOSE⁻ with the measured $R \leq \frac{3}{2} < 4$ and $L_1 \leq \frac{3}{5} < 3$, this implies $q_N < 1$ on those rungs. The constant $\frac{1}{2}$ is tight at $k=3$ (0.4868) and is not scramble-stable (seed 1 at $k_z=9$ gives $|Z_{\text{loc}}| = 0.524$).*

Cofinal $k \rightarrow \infty$ is not proved: $k=8$ is pinned, $k=10$ border-fails (r421), and $\frac{1}{2}$ is a finite-list ceiling.

6 Kills

World	Four coordinates	Role
χ_3 15, 23; χ_4 20	$q_N > 1$, sch > 0 , $ Z_{\text{loc}} > M$; overlap $10^{-2} / 10^{-4} / 10^{-4}$	named death
scramble seed 1 at $k_z=9$	$q_N = 0.270$ living; $ Z_{\text{loc}} = 0.524 > \frac{1}{2}$	kills $Z0_{\text{sel}}$ off MAIN
two-period $S=21$, $c=\frac{2}{3}$	$\text{nneg}(A_0) = 4$	chart dies
depth $n+1$ at $k_z=9$	$C_{\min}: 0.857 \rightarrow 1.000$	P1 razor is depth-sensitive
false anchor $k_z=16$	$q_N = 0.9805$ living, $ Z_{\text{loc}} = 0.756$, overlap $5 \cdot 10^{-9}$	42-rung razor; selected skips it
42-rung regression	$q_N \in [0.0015, 0.9805]$, 0/42 above 1	old min-margin 0.0195 reproduces

7 What this does not do

This is a finite selected census plus a named refutation of global overlap-equivalence. It does not prove cofinal $q_N < 1$, does not construct k_0 , and does not move the mincut (base 4 / refined 5). The $k=8$ window is a builder pin, not a live four-coordinate row. No L^* claim, no $R^\dagger$ claim, no RH claim.

References

- [1] Round 378, `compose_lemma`.
- [2] Round 386, `compose_premises2`.
- [3] Round 410, `hole_nyquist`.
- [4] Round 417, `source_sch_sign`.
- [5] Round 421, `reserve_limit`.

[6] Round 422, `sigma_limit`.

[7] Round 425, `cross_chain_gamma`.